

VG11 – HUBBLE TENSION AND NETWORK CLOSURE: FROM ANOMALY TO VG SIGNATURE

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Abstract

The Hubble Tension is one of the greatest anomalies in modern cosmology: local measurements (Cepheids, supernovae) and those from the CMB (Cosmic Microwave Background) yield two incompatible values for the Hubble constant. VG11 proposes an explanation based on **network dynamics** – not on a measurement error or an exotic particle, but on a **phase transition in the average coupling** of the Primary Network. Thus, what appears as an anomaly becomes a **signature** of how the Network relaxes over cosmic time. The episode connects these dynamics to the **closure process** of the Network, showing that accelerated expansion is only one phase of a larger cycle.

Keywords: Hubble Tension, average coupling J_{med} , phase transition, network relaxation, closure, VG.

1. INTRODUCTION – WHAT IS THE HUBBLE TENSION?

The Hubble constant (H_0) measures the rate of expansion of the universe. Two independent methods yield two values that do not match:

- **Local method (Cepheids, supernovae)** → ~ 74 km/s/Mpc
- **CMB method** → ~ 67 km/s/Mpc

The discrepancy of ~ 7 km/s/Mpc is too large to be explained by measurement errors. It is an **anomaly** indicating that the standard model (Λ CDM) is incomplete.

2. THE DEEP EQUATION – A GIFT FROM DEEP1

$$H(t) = -\frac{1}{\alpha} \cdot \frac{\dot{J}_{med}(t)}{J_{med}(t)}$$

Meaning of the terms:

- **$H(t)$** = **Hubble constant** at cosmic time t (how fast the universe is expanding now).
- **$J_{med}(t)$** = **average coupling** of the network at time t (how tightly the nodes are linked).
- **$\dot{J}_{med}(t)$** = **rate of change** of the average coupling at time t . The minus sign indicates that when the coupling decreases ($\dot{J}_{med} < 0$), expansion accelerates.
- **α** = a **dimensionless constant** linking geometry to network dynamics.

Physical interpretation:

Accelerated expansion is the **relaxation of the average coupling** of the Network. When J_{med} decreases rapidly, H is large. When J_{med} stabilizes, H decreases.

Important note:

This equation holds for any phase of the cycle – both expansion and contraction. The sign of $H(t)$ changes with the sign of $\dot{J}_{med}(t)$.

3. THE TWO VG SCENARIOS FOR THE HUBBLE TENSION

3.1. **Phase transition in coupling** : $J_{med}(t)$ underwent a smooth phase transition in the last ~5 billion years, temporarily accelerating the rate of coupling decrease. This transition may be caused by:

- Matter density dropping below a certain threshold.
- A change in the dominant type of excitations in the network (from matter-bound to "freer" excitations).

3.2. **Spin filter reaction** : As the universe cooled, excitations activated their spin filters, reducing the global coupling efficiency and accelerating expansion. This effect is inherent to network dynamics and can be tested through correlations between $H(z)$ and spin properties.

4. LINK TO NETWORK CLOSURE

The Hubble Tension is not an isolated anomaly – it is a **manifestation of a broader process: network relaxation before closure**.

4.1. How does it relate to closure?

- **During expansion**, J_{med} decreases, and H is large. The universe dilates.
- **At the critical moment** (reversal threshold), J_{med} reaches a minimum, and H tends toward zero. Expansion stops.
- **During contraction**, J_{med} begins to increase, and H becomes **negative** (indicating contraction). The universe shrinks back.

4.2. Signature of closure in current data

If we are approaching the reversal threshold, then:

- The decrease of H should be **slower** than in the Λ CDM model.
- There should be **small oscillations** in $H(z)$ at high redshifts ($z > 2$), which could be detected by future space missions.

4.3. Consequence for closure:

The Hubble Tension is an **indication** that we are approaching the **reversal threshold**. If this hypothesis is correct, then:

- Expansion will slow down over the next few billion years.
- Then, contraction will begin.

5. TESTABLE PREDICTIONS

- **Deviations from the Λ CDM curve** at different redshifts ($H(z)$), visible in distant supernova data.
- An extra "**shoulder**" or "**slope**" in the coupling transition epoch ($z \approx 0.5 - 1.5$).
- **Correlations between $H(z)$ and dark matter properties** (if dark matter is a nodal excitation).
- **Subtle oscillations** in $H(z)$ at high redshifts ($z > 2$) – a sign that the network is preparing for reversal.

6. CONCLUSION

The Hubble Tension becomes a **signature** of the Network's dynamics – a sign that accelerated expansion is the **relaxation of the average coupling** of the Network, not a mysterious dark energy. Moreover, it is an **indication** that we are approaching the **reversal threshold** – the moment when expansion slows down, stops, and contraction begins.

This interpretation links the Hubble Tension to the **closure process** of the Network, offering a unified perspective on the entire cosmic cycle: expansion $\rightarrow$ relaxation $\rightarrow$ reversal $\rightarrow$ contraction $\rightarrow$ rebirth.